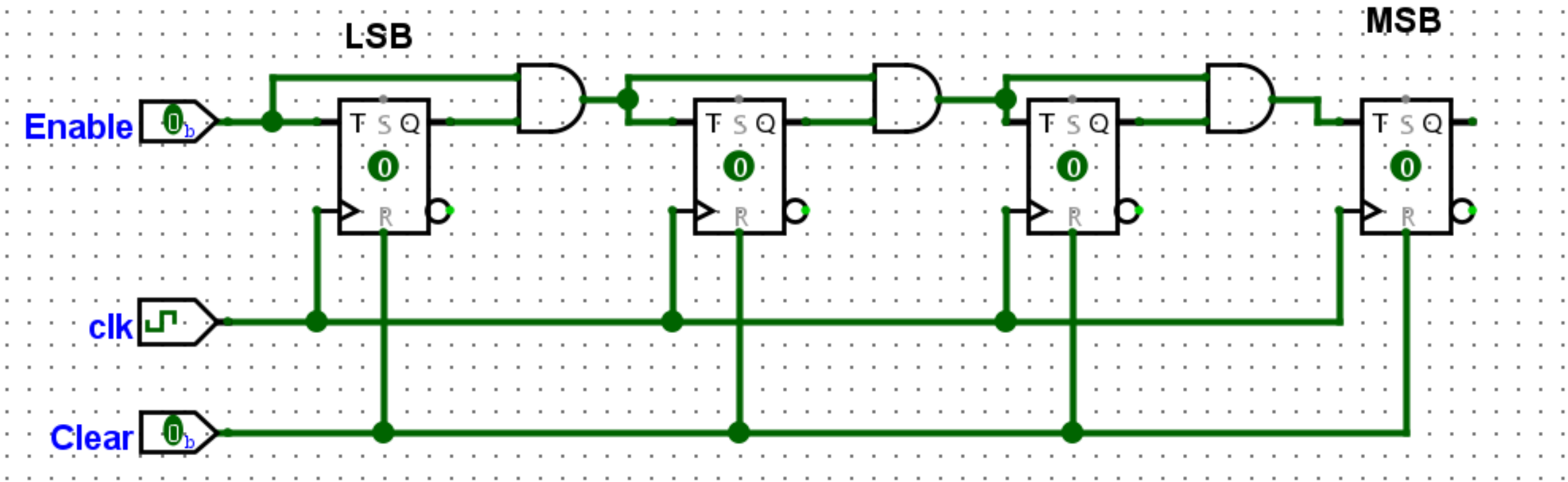
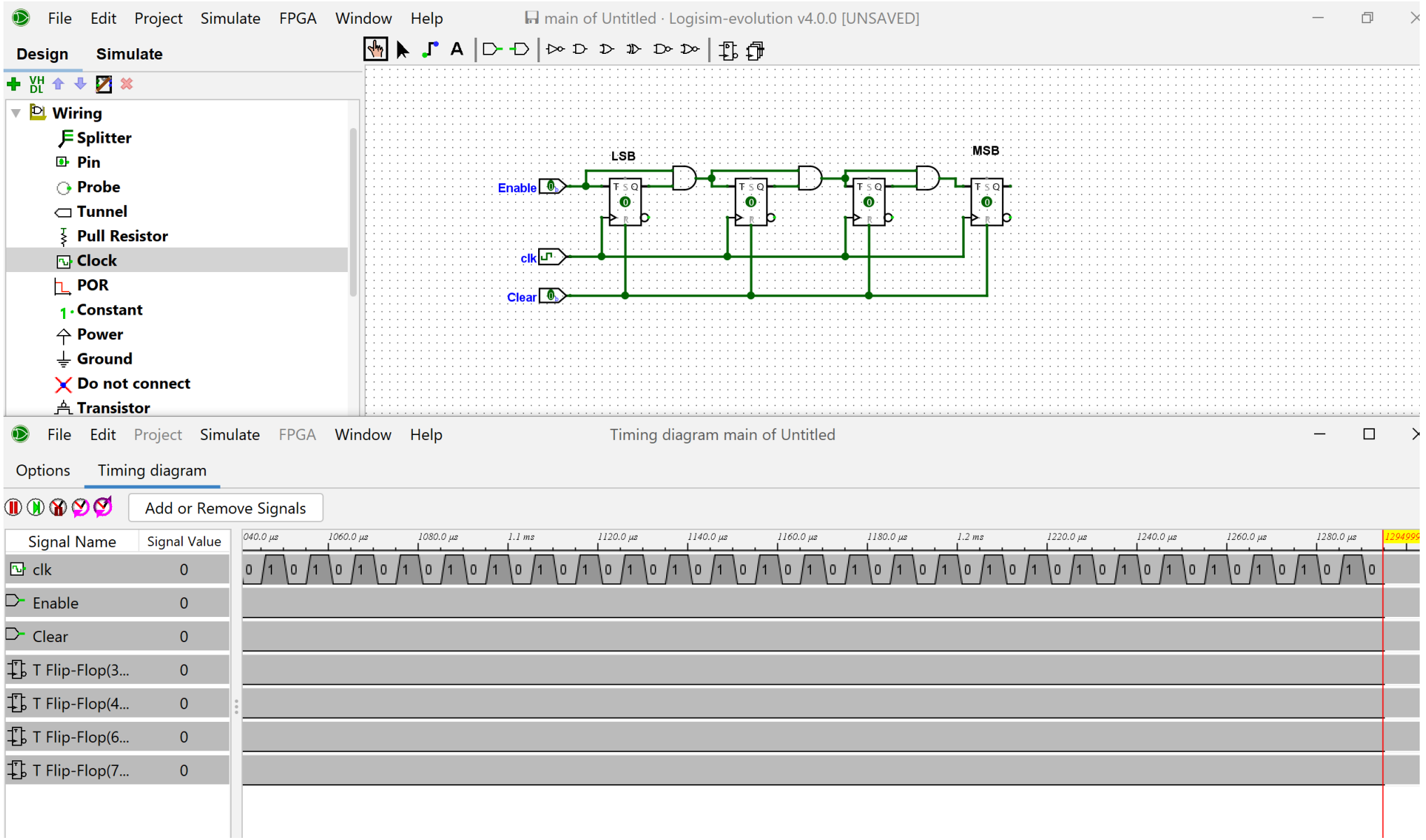
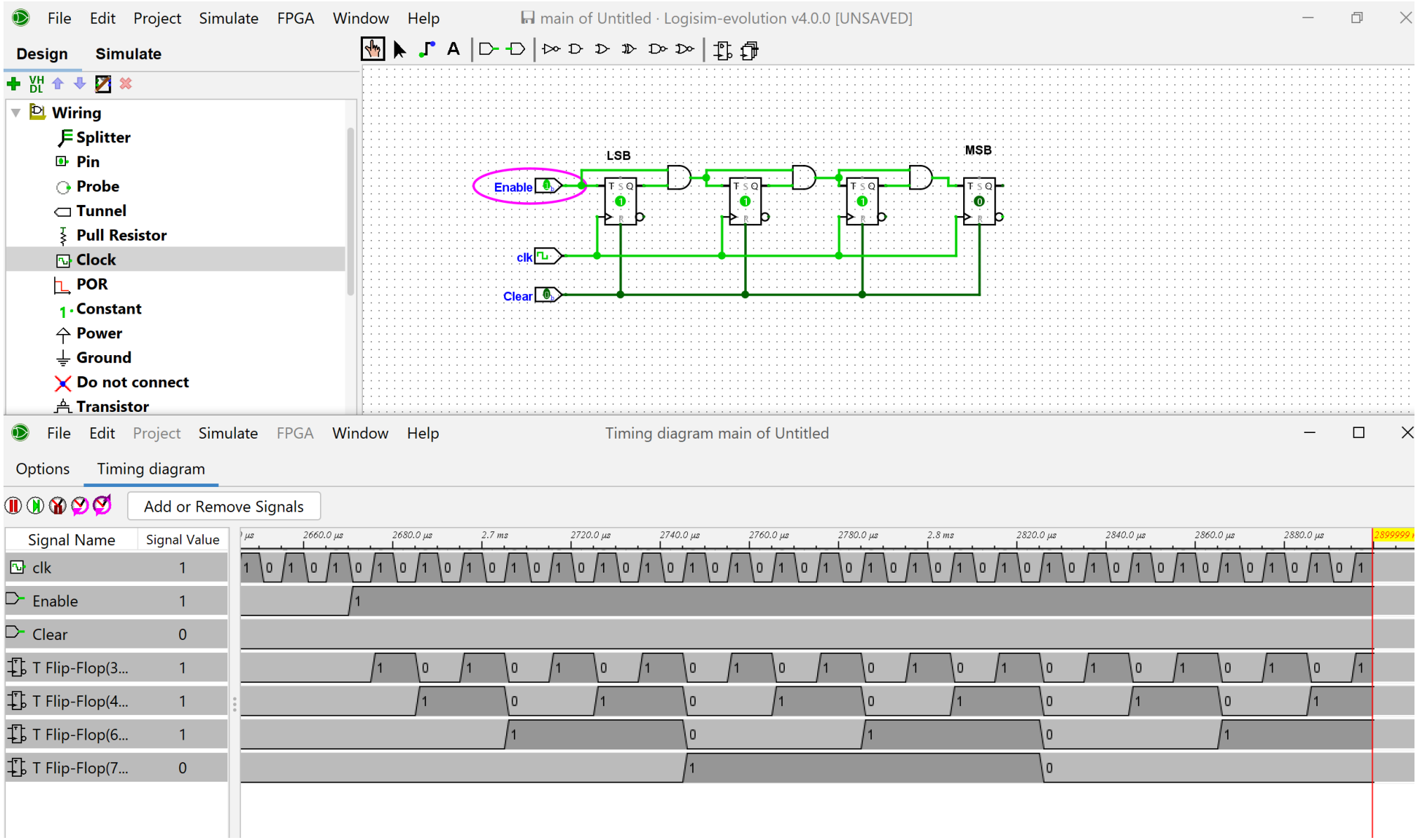
Part I: 4-Bit Counter Logisim Circuit and Time Diagrams



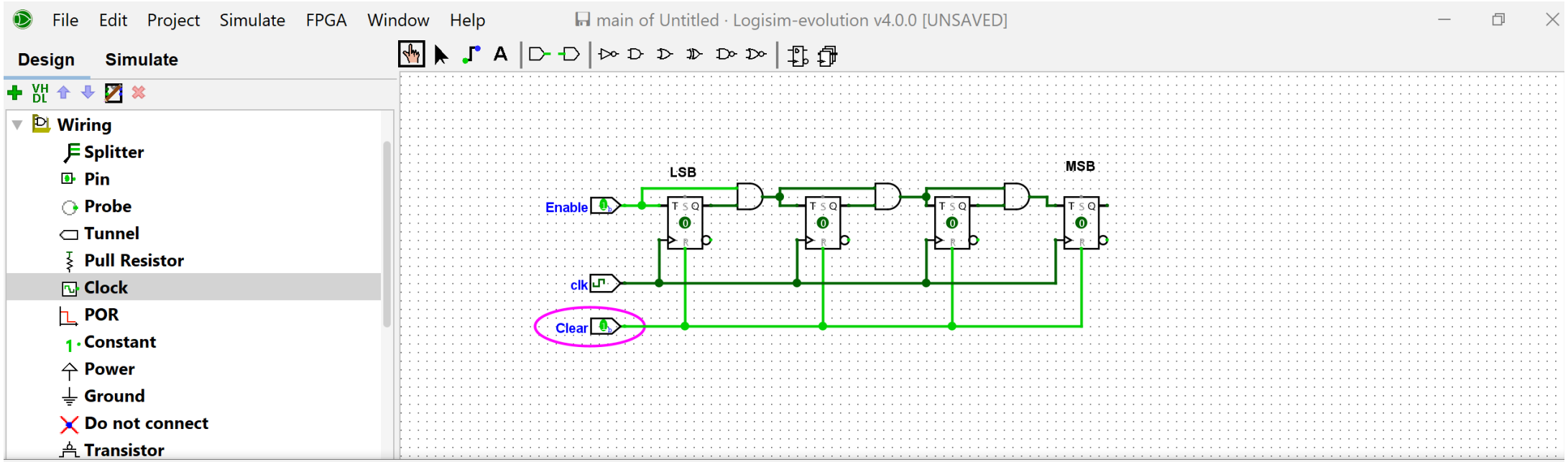
Part I: 4-Bit Counter Time Diagram (Only Clock Enabled)

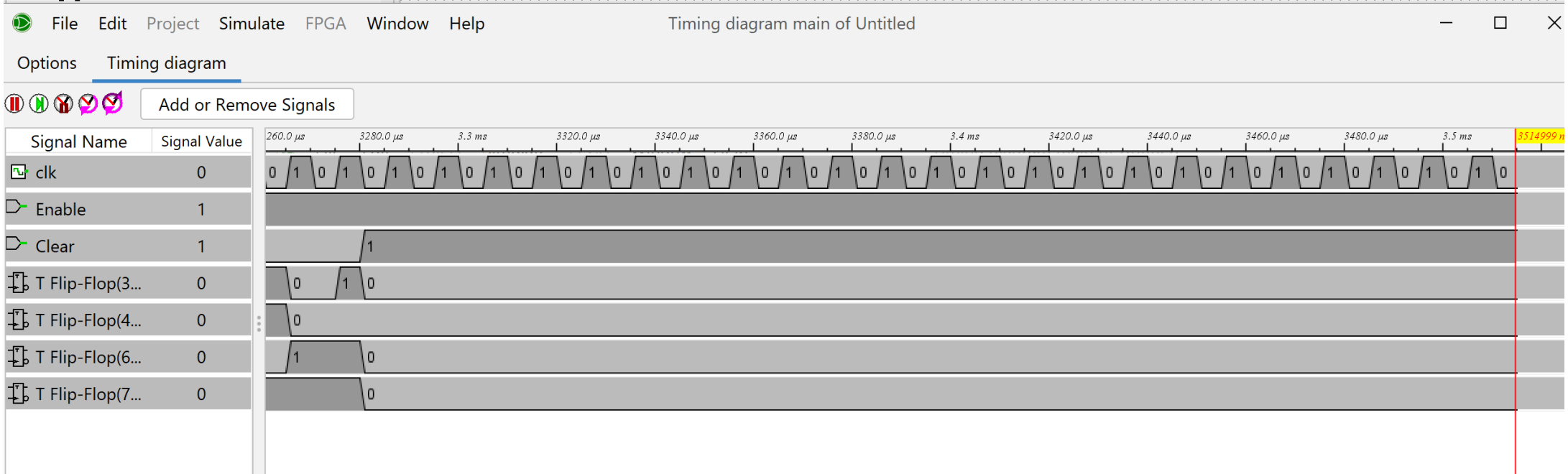


Part I: 4-Bit Counter Time Diagram (Clock and Enable Enabled)

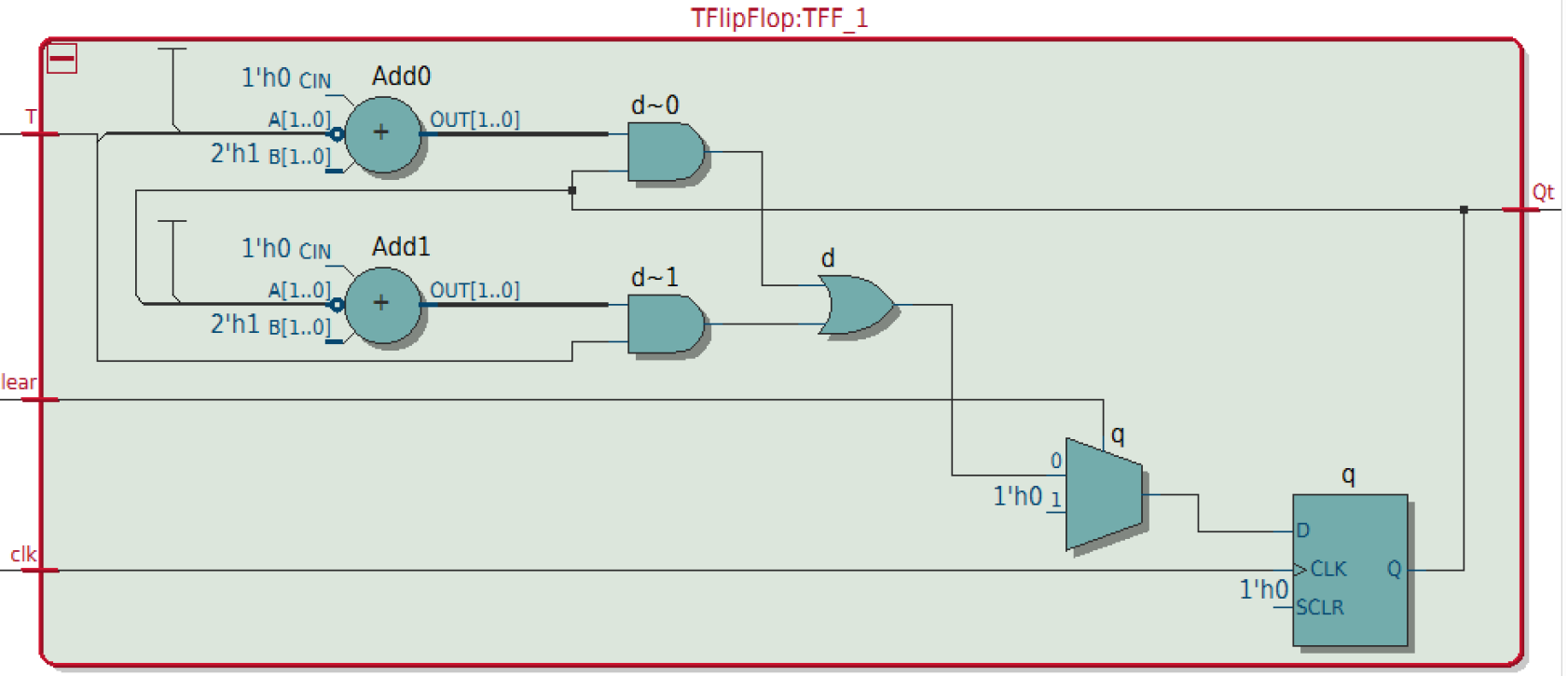


Part I: 4-Bit Counter Time Diagram (Clear Enabled)





Part I: RTL Viewer T Flip Flop




```
module TFlipFlop(
    input T,
    input clk,
    input clear,
    output Qt
);

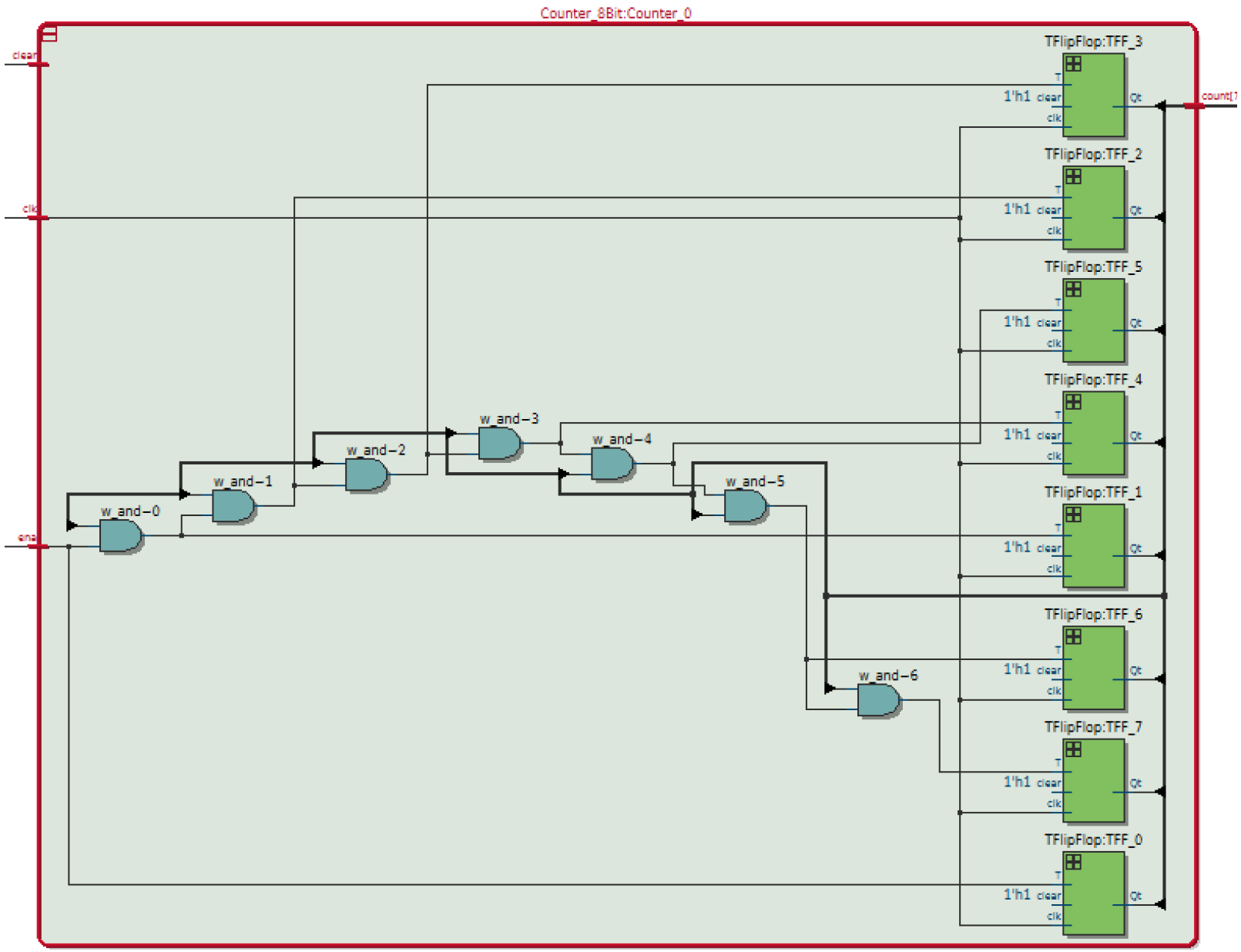
    wire d;
    reg q;

    // Edge Trigger Flip Flop
    always @(posedge clk) begin
        if (clear)
            q <= 0;
        else
            q <= d;
    end

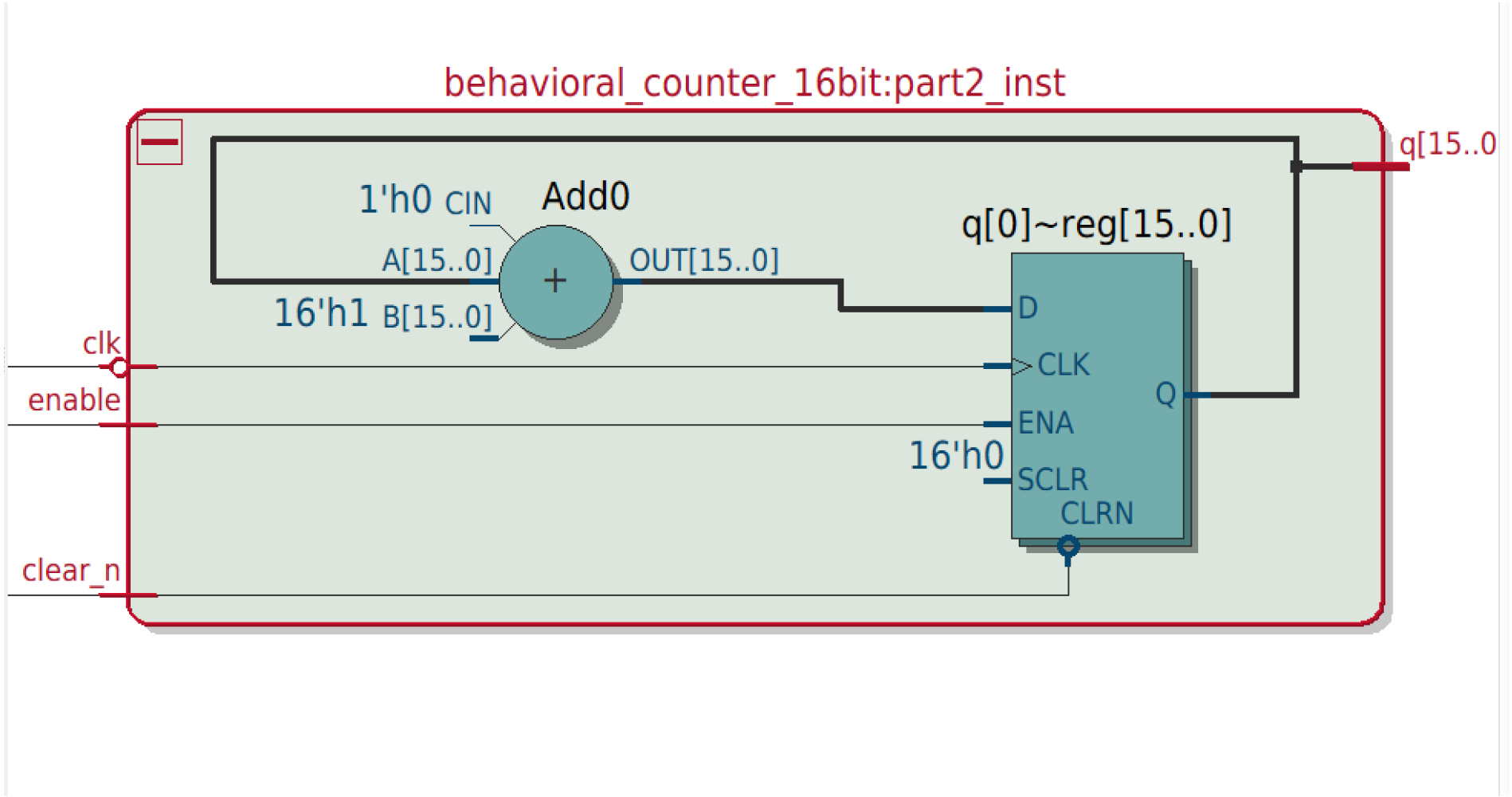
        assign d = (q & ~T) | (T & ~q);
        assign Qt = q;

endmodule
```

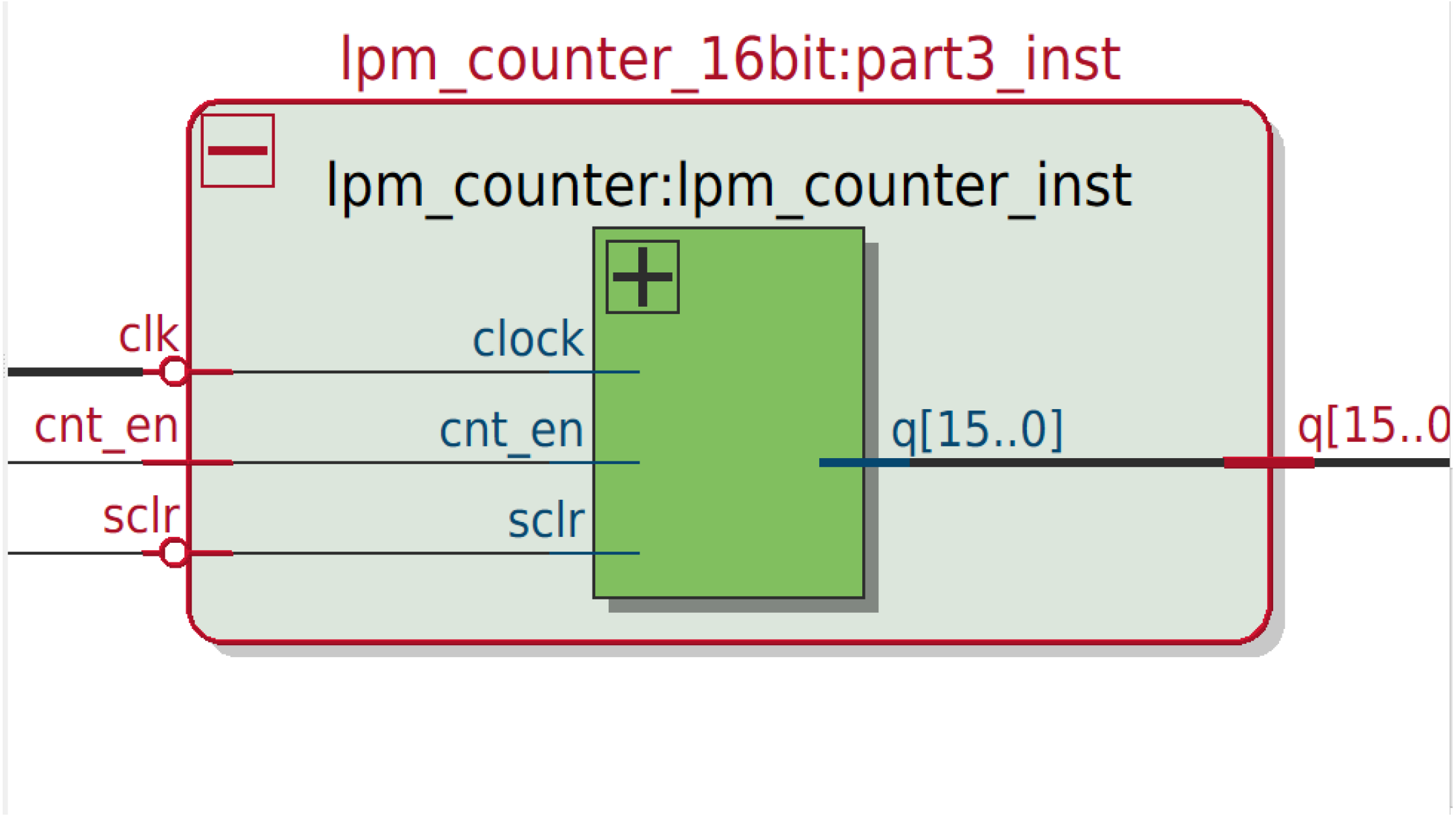
Part I: RTL Viewer 8Bit Counter



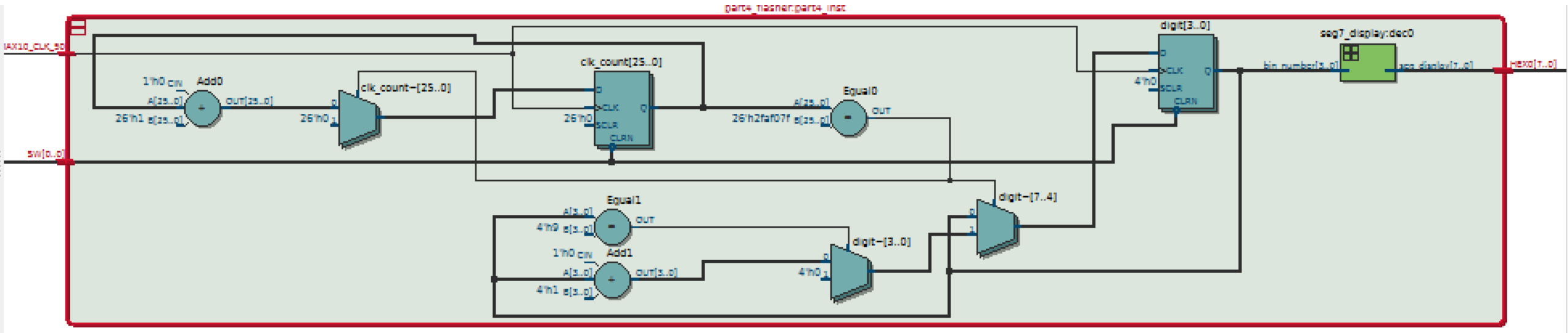
Part II: RTL Viewer behavioral counter 16bit



Part III: RTL Viewer lpm Counter 16bit



Part IV: RTL Viewer Flasher (0 through 9)



Part V: RTL Viewer "HELLO" Ticker

